University of Debrecen
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Design and Pick-and-Place Simulation of the RA-4 Robotic Arm

Robotic Arm Simulation Report

CodeAlpha Robotics & Automation Internship – Task 2

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1. Introduction

The aim of this task was to design a 3D model of a robotic arm, define its degrees of freedom (DOF) and perform a basic pick-and-place motion simulation. The arm, named RA-4, was modelled as an assembly in Autodesk Fusion, using joints with limits for every axis and the Motion Study tool for the simulation [2]. This note summarises the design, the kinematic structure and the simulated pick-and-place cycle. The CAD file (RA4_Arm.f3d), a STEP export and the screenshots are submitted with it.

2. Design of the RA-4

The RA-4 is a desktop arm intended for 3D printing and hobby servo actuation (Figure 1). A cylindrical base housing (Ø 140 mm) carries a top plate and a rotating turret. On the turret, the upper arm and the forearm form the shoulder and elbow, followed by a short wrist that carries the gripper. The arm joints have rectangular frames for servo motors. The distance between the shoulder and elbow axes is 150 mm, between the elbow and wrist axes 120 mm, and the fingertips are about 110 mm beyond the wrist axis, which gives a horizontal reach of approximately 380 mm from the base axis when the arm is fully stretched.

The gripper uses a rack-and-pinion mechanism. A pinion between the two fingers is connected to both finger sliders through motion links, so a single rotation opens or closes both fingers symmetrically, each by up to 20 mm. The model is placed on a 400 mm × 400 mm work table with a pick pad and a place pad, and a 20 mm cube serves as the workpiece. The upper arm carries the engraved text “RA-4 | M. Albusultan | 2026”. An interference analysis of the assembly showed no clashes between the parts.

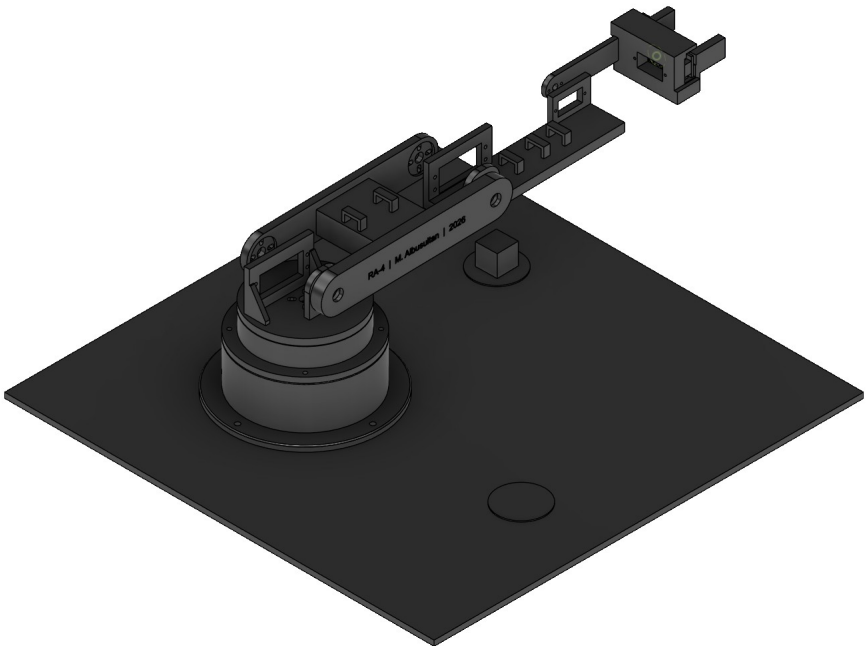


Figure 1. RA-4 robotic arm on the work table (home pose)

3. Degrees of Freedom

The arm has **four degrees of freedom**, all realised as revolute joints, plus a gripper (Table 1, Figure 2). J1 rotates the whole arm about the vertical axis of the base. J2, J3 and J4 have parallel, horizontal axes, so after J1 has chosen the direction, the arm moves in a vertical plane: J2 and J3 position the wrist in reach and height, and J4 sets the angle of the gripper. The gripper is the end

effector; it holds the part but does not change the position of the hand, so it is not counted as an arm DOF. The configuration is therefore described as 4 DOF + gripper.

Table 1. Degrees of freedom of the RA-4

DOF	Joint	Type / axis	Range in the model	Function
1	J1 – base	Revolute, vertical	−135° to +135°	Turns the arm towards the target
2	J2 – shoulder	Revolute, horizontal	−30° to +150°	Raises and lowers the upper arm
3	J3 – elbow	Revolute, horizontal	−150° to +30°	Sets reach and height
4	J4 – wrist	Revolute, horizontal	−90° to +90°	Keeps the gripper pointing down
–	Gripper	2 sliders + pinion	0 to 20 mm per finger	Grasps and releases the part

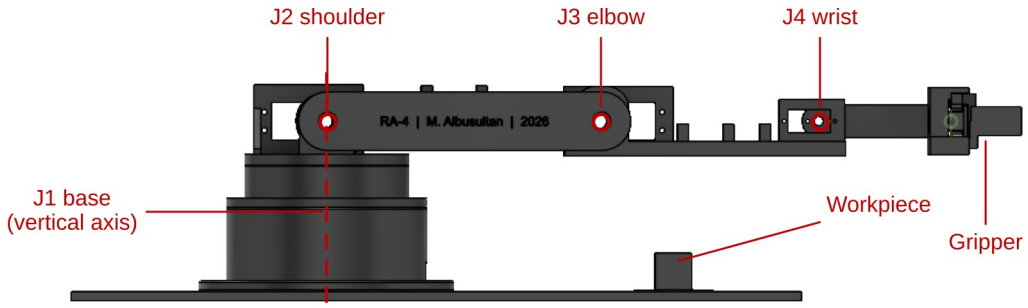


Figure 2. Side view of the RA-4 with the joint axes

Four DOF are sufficient for pick-and-place, where the part is approached from above and only its position changes. Because there is no wrist roll, the arm cannot rotate a part about the gripper axis; this would require a fifth joint, and full position and orientation control requires six. The ranges in Table 1 are those of the simulation model. The planned servos turn through 180°, so for the physical arm J1 will be limited to ±90°, while J2, J3 and J4 already span 180°.

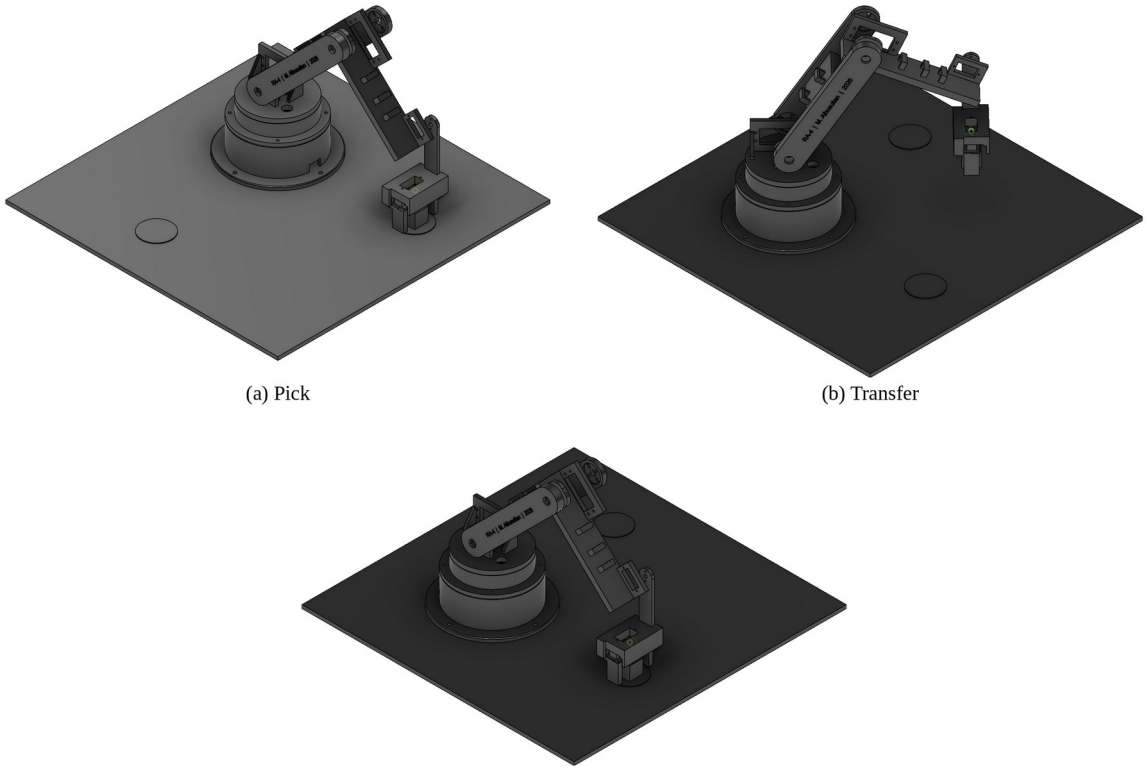
4. Pick-and-Place Simulation

The pick pad and the place pad are both 190 mm from the base axis and 90° apart, so the transfer is mainly a rotation of J1. The simulated cycle is: start in the home pose, move above the part, lower the gripper, close the fingers, lift the part, rotate the base by 90°, lower the part onto the place pad, open the fingers and return to home. In the motion study, the cube is connected to the base through a cylindrical joint about the base axis, so that it can rotate and rise together with the gripper during the transfer.

The joint angles of the key poses (Table 2) were calculated with the two-link inverse kinematics of J2 and J3 [1]. For a wrist target at horizontal distance r and height z relative to the shoulder, with $d^2 = r^2 + z^2$, the elbow angle follows from $\cos \theta 3 = (d^2 - L1^2 - L2^2) / (2\, L1\, L2)$, with L1 = 150 mm and L2 = 120 mm. J4 is then chosen so that J2 + J3 − J4 = −90°, which keeps the gripper vertical. Figure 3 shows the arm in these poses; the full cycle is shown in the motion study and the accompanying video.

Table 2. Joint angles of the key poses

Pose	J1	J2	J3	J4	Part
Home	0°	0°	0°	0°	On pick pad
(a) Pick	−0.2°	44.9°	−90.6°	44.3°	Gripped
(b) Transfer	45.0°	57.9°	−81.1°	66.9°	Lifted 60 mm
(c) Place	89.8°	44.9°	−90.6°	44.3°	On place pad



(a) Pick (b) Transfer

(c) Place

Figure 3. Key poses of the pick-and-place cycle

5. Conclusion

The RA-4 was designed in Autodesk Fusion as a 4-DOF articulated arm with a rack-and-pinion gripper. All joints are defined with limits, and the pick-and-place motion study shows that the arm can pick a 20 mm cube from one pad and place it on another 90° away, with the gripper kept vertical. All key poses lie within the 180° range of the planned servos. The next steps are printing the parts, mounting the servos and programming the same motion on a microcontroller.

References

[1] J. J. Craig, *Introduction to Robotics: Mechanics and Control*, 4th ed. Harlow, UK: Pearson, 2018.

[2] Autodesk, “Fusion help: Joints, motion links and motion studies.” [Online]. Available: https://help.autodesk.com/view/fusion360/ENU/